JOB & BATCH COSTING - CONCEPTS

1. Job Costing

- It is that form of specific order costing under which each job is treated as a cost unit and costs are accumulated and ascertained separately for each job.
- In other words, it is that form of specific order costing which applies where work is undertaken according to customer's requirement.
- It is generally used in industries where production is not on continuous basis, rather it is only when order from customers are received according to their specifications e.g. printing press, repair shop, etc.
- In this method cost of each job is computed by preparing the Job Cost Sheet.

Question – 1

The following data presented by the supervisor of a factory for a Job.

	₹ per unit
Direct material	120
Direct wages @ ₹ 4 per hour	60
(Department A-4 hrs., B-7hrs, C-2hrs & D-2hrs)	
Chargeable Expenses / D. Exp.	20
Total	PC → 200

PC 200

OHs
A = 4×4×100% = 16
B = 7×4×75% = 21
C = 2×4×90% = 7.2
D = 2×4×85% = 6.8 — 51
GFC/NFC — 251
(+) S. Exp. (30%×251) — 75.3
TC — 326.3
(+) Pft. (407.88×20%) — 81.58
SP — 326.3÷80% — 407.88

Analysis of the Profit and Loss Account for the year ended 31st March 2019

		₹		₹
Material used		2,00,000	Sales	4,30,000
Direct Wages:				
Dept. A	12,000			
Dept. B	8,000			
Dept. C	10,000			
Dept. D	20,000	50,000		
Special Stores Items		6,000		
Overheads:				
Dept. A	12,000			
Dept. B	6,000			
Dept. C	9,000			
Dept. D	17,000	44,000		
Gross Profit c/d		1,30,000		
		4,30,000		4,30,000
Selling Expenses		90,000	Gross Profit b/d	1,30,000

OH RR → $\frac{OH}{DL\ Cost}$

Dept. A = $\frac{12000}{12000}$ ×100 = 100% of DL

Dept. B = $\frac{6000}{8000}$ ×100 = 75% of DL

Dept. C = $\frac{9000}{10000}$ ×100 = 90% of DL

Dept. D = $\frac{17000}{20000}$ ×100 = 85% of DL

Sell. Exp. = $\frac{90000}{3L}$ ×100 = 30% of NFC

NFC = 2L + 0.50L + 0.06L + 0.44L = 3L

Net Profit	40,000	
	1,30,000	1,30,000

It is also to be noted that average hourly rates for all the four departments are similar. Required:

(i) Prepare a Job Cost Sheet
(ii) Calculate the entire revised cost using the above figures as the base.
(iii) Add 20% profit on selling price to determine the selling price.

Solution

Working Notes:

Overhead recovery rate on overall basis:

Overhead recovery rate = $\frac{44,000}{\left(\frac{50,000}{4}\right)}$ = ₹ 3.52

Statement of calculation of recovery rates

Particulars	Working	Recovery Rate
Dept. A	$\frac{12,000}{\left(\frac{12,000}{4}\right)}$	₹ 4 per direct labour hour
Dept. B	$\frac{6,000}{\left(\frac{8,000}{4}\right)}$	₹ 3 per direct labour hour
Dept. C	$\frac{9,000}{\left(\frac{10,000}{4}\right)}$	₹ 3.60 per direct labour hour
Dept. D	$\frac{17,000}{\left(\frac{20,000}{4}\right)}$	₹ 3.40 per direct labour hour
Selling exp. As % of NFC	$\frac{90,000}{2,00,000 + 50,000 + 6,000 + 44,000} \times 100$	30% of NFC

Statement of calculation of cost and selling price of Job

Particulars	Working	Amount (₹)
Material		120
Wages		60
Chargeable expenses		20
Prime Cost		200
(+) Overheads	Dept. A = 4 × 4.00 = 16 Dept. B = 7 × 3.00 = 21 Dept. C = 2 × 3.60 = 7.20 Dept. D = 2 × 3.40 = 6.80	51

GFC\NFC		251
(+) Selling expenses	30% × 251	75.30
Total Cost		326.30
Add: Profit	407.88 × 20%	81.58
Selling price	326.30 ÷ 80%	407.88

Question – 2

In a manufacturing company, factory overheads are charged as fixed percentage basis on direct labour and office overheads are charged on the basis of percentage of factory cost. The following information are available related to the year ending 31st March:

	Product A	**Product B**
Direct materials	₹ 19,000	₹ 15,000
Direct Labour	₹ 15,000	₹ 25,000
Sales	₹ 60,000	₹ 80,000
Profit	25% on cost	25% on sales price

You are required to find out:

(a) The percentage of factory overheads on direct labour x %

(b) The percentage of office overheads on factory cost y %

Solution

Let factory OH % on Direct labour = x

Let administration OH % on net factory cost = y

Statement of Cost

	Product A	**Product B**
Sales	60,000	80,000
Profit	12,000 [(60,000 × (25/125)]	20,000 (80,000 × 25%)
Total Cost	48,000	60,000

Statement of Cost

	Product A	**Product B**
Direct Material	19,000	15,000
Direct labour	15,000	25,000
Prime cost	34,000	40,000
Factory OHs	150x	250x
NFC/COP/COGS	34,000 + 150x	40,000 + 250x
(+) Admin. OH	340y + 1.5xy	400y + 2.5xy
COS	34,000 + 150x +340y + 1.5xy	40,000 + 250x + 400y + 2.5xy

$\therefore 34{,}000 + 150x + 340y + 1.5xy = 48{,}000$ ________(1)

& $40{,}000 + 250x + 400y + 2.5xy = 60{,}000$ ________(2)

Multiply equation (1) by 2.5 and equation (2) by 1.5 and subtract them, we get

$85{,}000 + 375x + 850y + 3.75xy = 1{,}20{,}000$

$\pm 60{,}000 \pm 375x \pm 600y \pm 3.75xy = \pm 90{,}000$

We get,

$\Rightarrow 25{,}000 + 250y = 30{,}000$

$\boxed{y = 20}$

Put value of y = 20 in equation (1),

$34{,}000 + 150x + 340(20) + 1.5x(20) = 48{,}000$

$\boxed{x = 40}$

Thus, Factory OH % on direct labour = 40% and administration OH % on factory cost = 20%

Question – 3	

In a factory following the Job Costing Method, an abstract from the work-in-progress as at 30th June was prepared as under:

OH RR	**Job No.**	**Materials ₹**	**Direct Labour Hours**	**Labour ₹**	**Factory overheads applied**
640/800 = 80%	115	0 + 1,325 = 1325	400 hrs	125 + 800 = 925	640 + 100 = 740
400/500 = 80%	118	515 + 810 = 1325	250 hrs	330 + 500 = 830	400 + 264 = 664
380/475 = 80%	120	665 + 765 = 1430	237.5 hrs	245 + 475 = 720	380 + 196 = 576
		₹ 2,900		₹ 1,775	₹ 1,420

Materials used in July were as follows:

Material requisition no.	**Job No.**	**Cost (₹)**
~~54~~ Return	118	300
55	~~118~~ 124	425
56	118	515 ✓
57	120	665
58	121	910
59	124	720
		3,535

	115	118	120
Mat.	1325	1325	1430
Lab.	925	830	720
F.OH	740	664	576
Fac. Cost	2990	2819	2726
(+) A. & S @ 10%	299	282	273
TC	3289	3101	2999
(+) Profit @ 20%	658	620	600
SP	3947	3721	3599

A summary of labour hours deployed during July is as under:

Job No.	**Number of hours**	
	Shop A @ Rs. 3	**Shop B** @ Rs. 2
115	25 × 3 = 75	25 × 2 = 50
118	90 × 3 = 270	30 × 2 = 60
120	75 × 3 = 225	10 × 2 = 20
121	65	--
124	20	10
	275	75

Indirect labour:	Waiting for material	20	10
	Machine Breakdown	10	5
	Idle Time	5	6
	Overtime Premium	6	5
		316	101

A shop credit slip was issued in July that material issued under Requisition No. 54 was returned back to stores as being not suitable. A material transfer note issued in July indicated that material issued under Requisition No. 55 for Job 118 was directed to Job 124.

The hourly rate in shop A per labour hour is ₹ 3 per hour while at shop B, it is ₹ 2 per hour. The factory overhead is supplied at the same rate as in June. Job 115, 118 and 120 were completed in July.

You are asked to compute the factory cost of the completed jobs. It is the practice of the management to put a 10% on the factory cost to cover administration and selling overheads and invoice the job to the customer on a total cost plus 20% basis. What would be the invoice price of these jobs?

Solution

Factory cost statement for completed jobs

Month	Job No.	Material	Direct labour	Factory OHs	Factory cost
September	115	1,325	800	640	2,765
October	115	-	125	100	225
Total		1,325	925	740	2,990
September	118	810	500	400	1,710
October	118	515	330	264	1,109
Total		1,325	830	664	2,819
September	120	765	475	380	1,620
October	120	665	245	196	1,106
Total		1,430	720	576	2,726

Invoice price of job

Job No.	115 (₹)	118 (₹)	120 (₹)
Factory cost	2,990.00	2,819.00	2,726.00
Administration and selling OHs @ 10% of factory cost	299.00	281.90	272.60
Total cost	3,289.00	3,100.90	2,998.60
Profit (20% of total cost)	657.80	620.18	599.72
Invoice Price	3,946.80	3,721.08	3,598.32

Indirect labour costs have been included in the factory overhead which has been recovered as 80% of the labour cost.

2. Batch Costing

- It is that form of specific order costing which applies where similar articles are manufactured in batches either for sale or use within the undertaking.
- Each batch of output is a cost unit and is costed separately.
- The total batch cost divided by number of units produced in a batch gives cost per unit.
- It is generally undertaken in case of pharmaceutical production, shoes, garments, etc.

3. Economic Batch Quantity (EOQ)

It is that batch size at which sum total of ordering cost and carrying cost is minimum.

$$EOQ = \sqrt{\frac{2 \times A \times S}{C}}$$

Where, A = Annual requirement of raw material

S = Set-up cost per batch

C = Carrying cost per unit per annum

$$\text{Number of set-up} = \frac{A}{\text{Batch Quantity}}$$ (Always round off to next complete value)

Total Set-up cost = No. of setup × Cost per set-up

$$\text{Average quantity} = \frac{\text{Batch quantity}}{2}$$

Total carrying cost = Average quantity × Carrying cost per unit per annum

Question – 4	

AUX ltd. has an annual demand from a single customer for 60,000 Covid-19 Vaccines. The customer prefers to order in the lot of 15,000 vaccines per order. The production cost of vaccine is ₹ 5,000 per vaccine. The set-up cost per production run of Covid-19 vaccines is ₹ 4,800. The carrying cost is ₹ 12 per vaccine per month.

$$\sqrt{\frac{2 \times 60000 \times 4800}{(12 \times 12)}} = 2000$$

You are required to:

(i) Find the most Economical Production Run

(ii) Calculate the extra cost that company incurs due to production of 15,000 vaccines in a batch.

Solution

(i) Annual demand = A = 60,000 vaccines

Set-up cost per run = S = ₹ 4,800

Carrying cost per unit per annum = C = ₹ 12 × 12 = ₹ 144

$$\text{Economic Batch Quantity} = \sqrt{\frac{2 \times A \times S}{C}} = \sqrt{\frac{2 \times 60{,}000 \times 4{,}800}{144}} = 2{,}000 \text{ vaccines}$$

	2000	15000
No. of Set-up	$\frac{60000}{2000} = 30$	$\frac{60000}{15000} = 4$
Set-up Cost	30×4800 = 1.44 l	4×4800 = 19200
CC	$\frac{2000}{2} \times 12 \times 12 = 1.44$ l	$\frac{15000}{2} \times 12 \times 12 = 10.8$ l
TC	2.88 l	1099200

Saving = 811200

(ii) **Statement of Cost**

Particulars	Batch size = 2,000 vaccines	Batch size = 15,000 vaccines
Set-up cost	$\frac{60,000}{2,000} \times 4,800 = 1,44,000$	$\frac{60,000}{15,000} \times 4,800 = 19,200$
Carrying cost	$\frac{2,000}{2} \times 144 = 1,44,000$	$\frac{15,000}{2} \times 144 = 10,80,000$
Total Cost	2,88,000	10,99,200

Extra cost = ₹ 10,99,200 – ₹ 2,88,000 = ₹ 8,11,200

Question – 5

A jobbing factory has undertaken to supply 300 pieces of a component per month for the ensuing six months. Every month a batch order is opened against which materials and labour hours are booked at actual. Overheads are levied at a rate per labour hour. The selling price contracted for is ₹ 8 per piece. From the following data calculate the cost and profit per piece of each batch order and overall position of the order for 1,800 pieces.

Month	Batch output (Sales)	Material cost (₹)	Direct wages (₹)	Direct labour hours (OHs)	TC	P/L
January	8× 310 = 2480	1150	120	2.5× 240 = 600	1870	610
February	8× 300 = 2400	1140	140	2.4× 280 = 672	1952	448
March	8× 320 = 2560	1180	150	2.4× 280 = 672	2002	558
April	8× 280 = 2240	1130	140	2.3× 270 = 621	1891	349
May	8× 300 = 2400	1200	150	2.6× 300 = 780	2130	270
June	8× 320 = 2560	1220	160	2.5× 320 = 800	2180	380
	1830				12025	

The other details are:

Month	Chargeable expenses (₹)	Direct labour hours	RR
January	12,000	4,800	2.5
February	10,560	4,400	2.4
March	12,000	5,000	2.4
April	10,580	4,600	2.3
May	13,000	5,000	2.6
June	12,000	4,800	2.5
	70140 ÷	28600 =	2.4524

Sales = 1800×8 = 14400
(-) TC = $\frac{12025}{1830}$ × 1800 = 11828
P/L = 2572

Solution

Statement of Cost and Profit per batch

Particulars	Jan.	Feb.	March	April	May	June	Total
Batch output (in units)	310	300	320	280	300	320	1,830
Sale value (₹)	2,480	2,400	2,560	2,240	2,400	2,560	14,640
Material cost (₹)	1,150	1,140	1,180	1,130	1,200	1,220	7,020

Direct wages (₹)	120	140	150	140	150	160	860
Chargeable expenses* (₹)	589	687	687	662	736	785	4,146
Total cost (₹)	1,859	1,967	2,017	1,932	2,086	2,165	12,026
Profit per batch (₹)	621	433	543	308	314	395	2,614
Total cost per unit (₹)	6.00	6.56	6.30	6.90	6.95	6.77	6.57
Profit per unit (₹)	2.00	1.44	1.697	1.10	1.05	1.23	1.43

Overall position of the order for 1,200 units

Sales value of 1,800 units @ ₹ 8 per unit	₹ 14,400
Total cost of 1,800 units @ ₹ 6.57 per unit	₹ 11,826
Profit	₹ 2,574

*Chargeable Expenses Rate = $\frac{\textit{Total Chargeable Expenses}}{\textit{Total direct labour hours}} = \frac{70{,}140}{28{,}600}$ = ₹ 2.452448 per labour hour

It is assumed that recovery rate is based on overall 6 months period. Other way is to compute recovery rate for each month and then compute the cost.

Question – 6

Language Achievers, a renowned institute specializing in TOEFL preparation, has secured a spacious hall for 20,000 on weekly basis with a seating capacity of 250 students. The instructor, highly qualified and experienced, is compensated generously with an honorarium of 1,500 per lecture. Additionally, he receives reimbursement for travel expenses of ₹200 per day along with refreshments costing 1,500 per week to ensure his comfort and focus during teaching sessions. Administrative and miscellaneous expenses, covering essential utilities and materials are, 500 per week. Language Achievers has meticulously planned its curriculum, scheduling batches of 2 lectures per day, 5 days a week for 30 weeks, ensuring comprehensive coverage of the TOEFL syllabus.

Required:

(20000×30) + (1500×2×5×30) + (200×5×30) + (1500×30) + (500×30) = 11.40L

(i) Calculate the total cost per batch.

$\frac{11.40L}{250}$ = 4560

(ii) Determine the minimum fee per student in a batch to cover costs, if the batch is fully occupied.

(iii) Calculate the fee to be charged from. each student if batch is 80% filled and institute aims to achieve a profit margin of 25% on the fee.

Stud. = 250 × 80% = 200

Fee = $\frac{11.40L}{200} \times \frac{1}{75\%}$ = ₹7600

Solution

(i) Calculation of Total cost per batch

Particulars	Amount (₹)
Hall Charges (₹20,000 × 30)	6,00,000
Honorarium of instructor (₹1,500 × 2 × 5 × 30)	4,50,000
Reimbursement of travel expenses (₹200 × 5 × 30)	30,000
Refreshment (₹1,500 × 30)	45,000
Administrative and miscellaneous expenses (₹500 × 30)	15,000
Total Cost	11,40,000
No. of Batches	1
Total cost per batch	11,40,000

(ii) Minimum fee per student in a batch to cover costs

$$= \frac{\text{Total cost per batch}}{\text{Students per batch}} = \frac{11,40,000}{250} = ₹4,560$$

(iii) Number of Students if batch is 80% filled

= 250 students × 80% = 200 students

Total Fee to be recovered to achieve 25% profit margin on the fee

= ₹11,40,000 + (₹11,40,000 × 1/4th of sales or 1/3rd of the cost) = ₹15,20,000

$$\text{Fee per student} = \frac{\text{Total Fee per batch}}{\text{Students per batch}} = \frac{15,20,000}{200} = ₹7,600$$

Question – 7	

Pinku Confectioners (PC) owns a bakery which is used to make bakery items like pastries, cakes and muffins. PC use to bake atleast 50 units of any item at a time. A customer has given an order for 600 cakes. To process a batch of 50 cakes, the following cost would be incurred:

No. of Batch = $\frac{600}{50}$ = 12

Direct materials	- ₹5,000	→ 60000	$\frac{5000}{50}$ × 605 = 60500
Direct wages	- ₹500	→ 6000	$\frac{500}{50}$ × 605 = 6050
Oven set- up cost	- ₹750	→ 9000	750×13 = 9750

No. of Batch = $\frac{605}{50}$ = 12.1 ≈ 13

PC absorbs production overheads at a rate of 20% of direct wages cost. 10% is added to the total production cost of each batch to allow for selling, distribution and administration overheads. PC requires a profit margin of 25% of sales value.

Required:

(i) Determine the price to be charged for 600 cakes.

(ii) Calculate cost and selling price per cake.

(iii) What would be selling price per unit If the order is for 605 cakes.

$\frac{83820}{600}$ = 139.7

	75000		76300
A.OH (20% × 6000)	1200	(20% × 6050)	1210
	76200		77510
(+) 10%	7620		7751
	83820		85261

(+) P/F

Sale $\frac{83820}{75\%}$ = 111760 ÷ 600 = 186.27

$\frac{85261}{75\%}$ = 113681 ÷ 605 = 187.90

Solution

Statement of cost per batch and per order

No. of batch = 600 units ÷ 50 units = 12 batches

	Particulars	Cost per batch (₹)	Total Cost (₹)
	Direct Material Cost	5,000.00	60,000
	Direct Wages	500.00	6,000
	Oven set-up cost	750.00	9,000
	Add: Production Overheads (20% of Direct wages)	100.00	1,200
	Total Production cost	6,350.00	76,200
	Add: S&D and Administration overheads (10% of Total production cost)	635.00	7,620
	Total Cost	6,985.00	83,820
	Add: Profit (1/3rd of total cost)	2,328.33	27,940
(i)	**Sales price**	**9,313.33**	**1,11,760**
	No. of units in batch	50 units	
(ii)	**Cost per unit** (₹6,985 ÷ 50 units)	**139.70**	
	Selling price per unit (9,313.33 ÷ 50 units)	**186.27**	

(iii) If the order is for 605 cakes, then selling price per cake would be as below:

Particulars	Total Cost (₹)
Direct Material Cost	60,500
Direct Wages	6,050
Oven set-up cost	9,750
Add: Production Overheads (20% of Direct wages)	1,210
Total Production cost	77,510
Add: S&D and Administration overheads (10% of Total production cost)	7,751
Total Cost	85,261
Add: Profit (1/3rd of total cost)	28,420
Sales price	**1,13,681**
No. of units	605 units
Selling price per unit (₹ 1,13,681 ÷ 605 units)	**187.90**